

ECG Anomaly Detection — Session Report

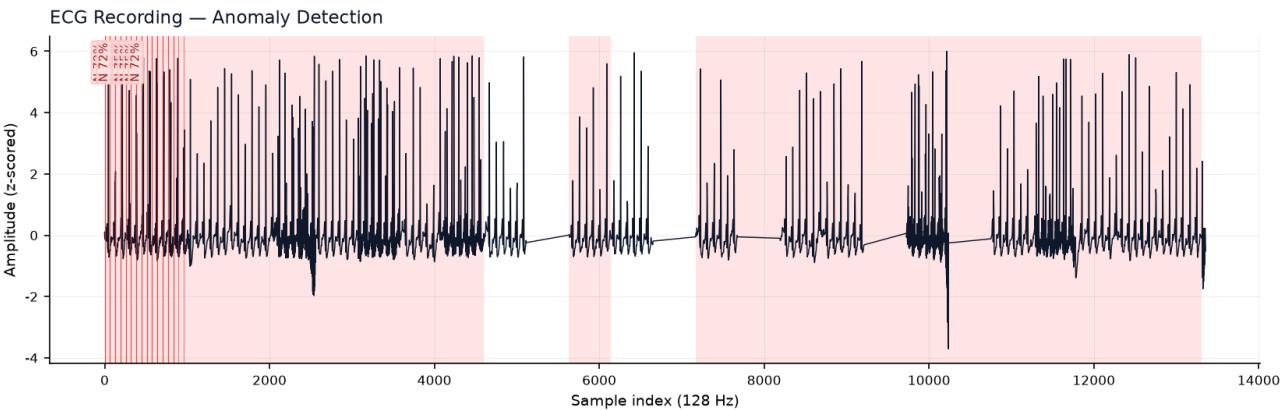
Prepared for: II • Session ID: e483ae07 • Generated: 2026-07-05 06:23:14

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (234)
Started	2026-07-05 06:23:14
Ended	2026-07-05 06:26:48
Duration	212.7 s
Status	completed
Total beats analyzed	108,032
Anomalous beats	73,258 (67.81%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



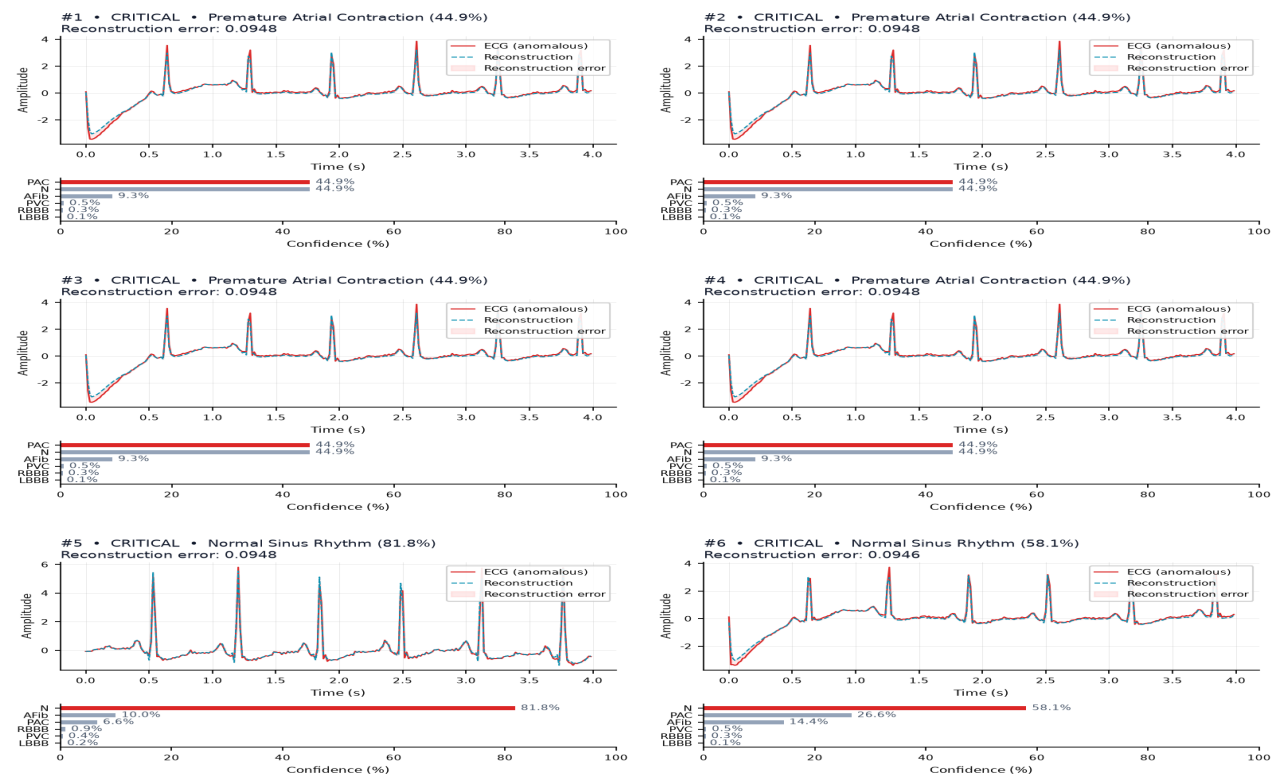
Main Arrhythmia Finding

Anomalous ECG regions were detected, but no arrhythmia classification reached the 50% confidence threshold. The report therefore does not name a main arrhythmia type; the detailed waveform examples below should be reviewed manually.

Detected Anomaly Waveforms

The following plots show example ECG segments (red) that triggered alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the

two curves represents the reconstruction error that exceeded the detection threshold. When available, examples are selected from arrhythmia classifications above 50% confidence.



Clinical Summary

67.8% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.